Table 1: Rating scale to assess the extent of fouling on inspection areas

Rating	Description	Macrofouling cover of area inspected (visual estimate)	Recommended cleaning
0	No fouling	Surface entirely clean. No visible biofouling on surfaces.	- -
1	Microfouling	Submerged areas partially or entirely covered in microfouling. Metal and painted surface may be visible beneath the fouling.	-	Proactive cleaning may be recommended as further specified in paragraph 9.4.
2	Light macrofouling	Presence of microfouling and multiple macrofouling patches. Fouling species cannot be easily wiped off by hand.	1-15% of surface	Cleaning with capture is recommended as further specified in paragraph 9.9.

It is recommended to shorten the

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SOURCE: MEPC.378(80).pdf

visible biofouling on surfaces.
- -
1 Microfouling
Submerged areas partially or entirely covered in microfouling. Metal and painted surface may be visible beneath the fouling.
-
Proactive cleaning may be recommended as further specified in paragraph 9.4.
2 Light macrofouling
Presence of microfouling and multiple macrofouling patches. Fouling species cannot be easily wiped off by hand.
1-15% of surface

Cleaning with capture is recommended as further specified in paragraph 9.9.

It is recommended to shorten the

interval until the next inspection.
If the AFS is significantly deteriorated, dry -docking with maintenance and reapplication of the AFS is recommended.
3 Medium macrofouling
Presence of microfouling and multiple macrofouling patches.
16-40% of surface
4 Heavy macrofouling
Large patches or submerged areas entirely covered in macrofouling.
41-100% of surface

Condition of the AFS

8.11 The condition of the AFS on the hull and in niche areas should be observed during the inspection and reported. Recommended action and relevant procedures for inspection of the AFS are described in tables 4 and 5.

Inspection report

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SOURCE: MEPC.378(80).pdf
8.11 The condition of the AFS on the hull and in niche areas should be observed during the inspection and reported. Recommended action and relevant procedures for inspection of the AFS are described in tables 4 and 5.

Inspection report

8.12 An inspection report should be prepared and a copy should be available on board and listed/linked in the Biofouling Record Book (BFRB). For details on reporting on biofouling levels and AFS condition inspections, see appendix 2, tables 4 to 6.

9 CLEANING AND MAINTENANCE

9.1 Cleaning is an important measure to remove biofouling from the hull and niche areas, but may physically damage the AFC, shorten coating service lifetime and release harmful waste substances and invasive aquatic species into the environment.

9.2 Comprehensive testing of cleaning systems or processes is necessary to understand the cleaning performance, capture efficiency or any release of harmful waste substances as well as improve knowledge concerning the prevention of release of viable fragments, spores and other parts of biofouling organisms that have the potential to be invasive.

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SOURCE: MEPC.378(80).pdf
well as improve knowledge concerning the prevention of release of viable fragments, spores and other parts of biofouling organisms that have the potential to be invasive.
RESOLUTION MEPC.378(80) (adopted on 7 July 2023)
2023 GUIDELINES FOR THE CONTROL AND MANAGEMENT OF SHIPS' BIOFOULING TO MINIMIZE THE TRANSFER OF INVASIVE AQUATIC SPECIESMEPC 80/17/Add.1
Annex 17, page 17

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9.3 In-water cleaning is a complex activity to manage appropriately and international standards for the management of in -water cleaning may continue to be developed and published in a stand-alone document to the Guidelines.

Procedures for proactive cleaning

9.4 Proactive cleaning is the periodic removal of microfouling on ships' hull and niche areas or other submerged surfaces as relevant prior to macrofouling growth and can be conducted with or without capture. Proactive cleaning without capture should:

.1 not be conducted on biofouling with rating 2 in line with table 1; and
.2 be performed in an area accepted by the relevant authority for this activity.

9.5 Operators undertaking proactive cleaning should be aware of any local regulations or

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SOURCE: MEPC.378(80).pdf
.1 not be conducted on biofouling with rating 2 in line with table 1; and
.2 be performed in an area accepted by the relevant authority for this activity.

9.5 Operators undertaking proactive cleaning should be aware of any local regulations or requirements. Regulations regarding the discharge of biofouling and waste substances into the marine environment and the location of sensitive areas (such as Marine Protected Areas) may be relevant.

9.6 Procedures for proactive cleaning and frequency should be described in the BFMP. All proactive cleaning, and any determination of biofouling level prior to the cleaning, should be entered in the BFRB.

Procedures for reactive cleaning

9.7 Reactive cleaning systems physically remove micro - and macrofouling from the hull and niche areas. There are various reactive cleaning methods available and more under development.

9.8 Reactive cleaning should be conducted based on the inspection results and contingency actions as outlined in table 1, though cleaning with capture may be used to manage any rating level.

9.9 The reactive cleaning should:

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SOURCE: MEPC.378(80).pdf
9.8 Reactive cleaning should be conducted based on the inspection results and contingency actions as outlined in table 1, though cleaning with capture may be used to manage any rating level.

9.9 The reactive cleaning should:

.1 use a reactive cleaning system that is compatible with the AFC in order to minimize damage of the AFC;

.2 be conducted with the aim of achieving a fouling rating 1 for the cleaned area in line with table 1;

.3 strive for effective collection and safe disposal of all biofouling material and waste substances when reactive cleaning is performed in water or at dry dock; and

.4 be performed in an area accepted by the relevant authority for this activity.

9.10 Biofouling management in niche areas should include the following or similar adequate measures:

.1 maintenance of any MGPS installed to ensure they operate effectively to